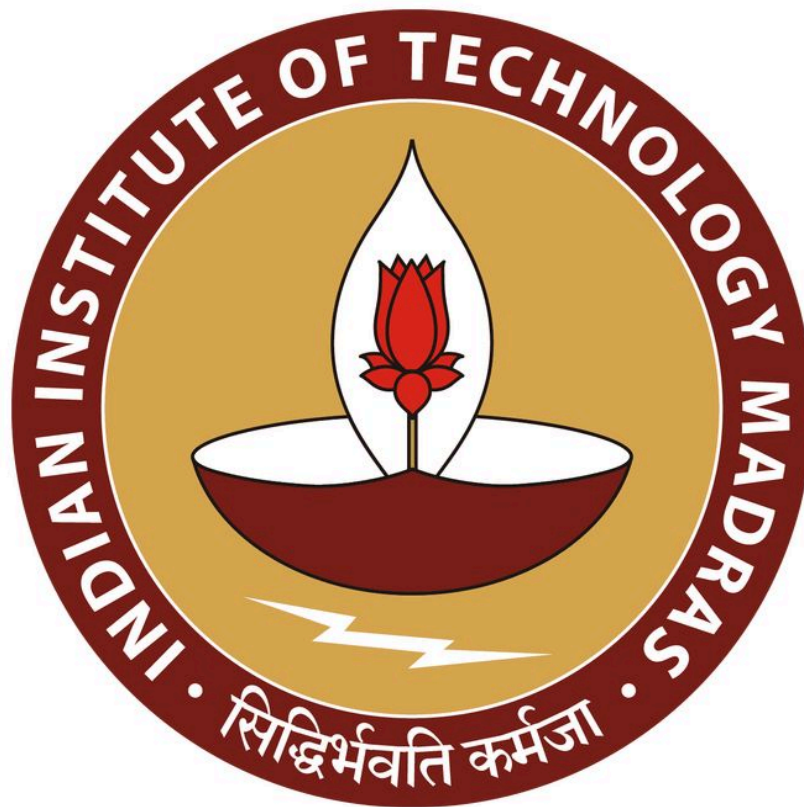


Inventory Optimization and Workforce Planning for Operational Sustainability: A Descriptive Business Analytics Study of Ashmiti Enterprise

An End-Term report for the BDM Capstone project

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1. Executive Summary

Introduction:

Ashmiti Enterprise is a Vadodara-based business engaged in printer management services, toner cartridge remanufacturing, IT infrastructure maintenance, equipment rentals, and Annual Maintenance Contract (AMC) services across multiple Indian states. With a diverse range of products and service operations, effective management of inventory, procurement, and workforce activities is important for maintaining operational efficiency and sustainable business performance.

Challenges:

The business faces operational challenges related to inventory management, fluctuating procurement requirements, concentration of capital in high-value inventory items, and workforce and service-operation variability. These challenges can affect inventory utilisation, working capital management, service reliability, and resource allocation. The absence of a structured analytical approach makes it difficult to identify priority inventory categories, understand procurement patterns, and evaluate workforce and service-operation performance systematically.

Key Findings: The analysis indicates that inventory value is highly concentrated, with the five highest-value categories accounting for 57.28% of classified inventory value, while A-class categories account for 69.55%. Procurement expenditure showed considerable month-to-month variation, whereas credit transaction activity was comparatively more stable. Service operations during the April–June 2026 analysis period recorded an 89.17% First-Time Fix Rate and a 91.94% complaint-free rate. SLA compliance was 93.59% for service visits associated with present employees compared with 45.83% for visits associated with unplanned employee absence. The procurement–credit transaction correlation was approximately 0.07, indicating very weak linear association.

2. Analysis Method and its Explanation

2.1 Data Collection

The study uses business records obtained from **Ashmiti Enterprise** for the period **January to June 2026**. The **Purchase Register**, **Credit Transaction Register**, and **Godown Summary** were extracted from the organisation's Tally ERP system and used to analyse procurement expenditure, business transaction activity, and inventory holdings, respectively. The Purchase Register and Credit Transaction Register contain six monthly observations each, while the Godown Summary contains 193 inventory-category records.

For inventory value and ABC analysis, only the 126 categories with positive inventory values were considered, as these represent the inventory currently contributing positively to the recorded inventory value. Records with zero or negative inventory values were excluded from the ABC classification. Such non-positive balances may arise from stock adjustments, returns, or other accounting entries and therefore were not treated as positive inventory holdings for the purpose of inventory concentration analysis.

Workforce and Service Operations Data: Workforce attendance and service-operation records were available for a three-month period from **April to June 2026**. The workforce dataset contains **936 employee-day attendance records covering 12 employees**, while the service-operations dataset contains **360 service-visit records** during the same period. These records were used to analyse workforce availability, attendance disruptions, service workload, SLA performance, service quality, and the relationship between workforce availability and service outcomes.

The available datasets were subsequently cleaned, structured, and analysed using Google Sheets. Links to the original input datasets and the complete analysis workbook are provided for review and reproducibility.

2.2 Data Cleaning and Pre-processing

The datasets obtained for the study were initially received in a compressed ZIP format and were extracted and converted into Excel-compatible files to facilitate further processing and analysis. The converted datasets were subsequently imported into Google Sheets, which was used as the primary tool for data cleaning, preparation, and analysis.

During preprocessing, date variables were formatted into a consistent date format to ensure uniformity across the records. Observations containing missing values in variables required for the analysis were removed to avoid incomplete records affecting the subsequent calculations. The datasets were then organised into structured tabular formats, and the relevant variables required for procurement, inventory, transaction, workforce, and service-operation analyses were retained.

The cleaned datasets were maintained separately from the source files to preserve data traceability. All subsequent calculations, summary tables, and visualisations were generated using the cleaned data in Google Sheets.

2.3 Quantitative Analysis Methods

2.3.1 Descriptive Statistical Analysis

Business problem: The available business records contain variations in procurement

expenditure, credit transaction activity, and inventory values. A statistical summary is required to establish a baseline understanding of these variables before conducting detailed analysis.

Data used: Purchase Register, Credit Transaction Register, and Godown Summary.

Method: Descriptive statistics including total, mean, median, minimum, maximum, range, and standard deviation were calculated for the relevant numerical variables. These measures were used to assess the central tendency and variability of procurement expenditure, credit transaction values, and inventory values.

Tool: Google Sheets.

Purpose: To summarise the available business data and identify notable variations and concentration patterns that require further investigation.

2.3.2 Procurement Trend and Variability Analysis

Business problem: Procurement expenditure varies across months, making it important to understand purchasing patterns and the extent of variation in procurement activity.

Data used: Purchase Register containing monthly procurement expenditure for January–June 2026.

Method: Monthly purchase expenditure was arranged chronologically and analysed using month-wise values, percentage contribution, and summary measures of variability. A column chart was used to visualise changes in procurement expenditure across the study period. The coefficient of variation was used to assess procurement variability relative to the average monthly expenditure.

Formulae Used: $\text{Mean} = \sum x_i / n$

$\text{SD} = \sqrt{[\sum (x_i - \bar{x})^2 / (n - 1)]}$

$\text{CV} = (\text{SD} / \text{Mean}) \times 100$

Tool: Google Sheets.

Purpose: To identify periods of relatively high and low procurement activity and provide a quantitative basis for evaluating procurement planning and inventory replenishment practices.

Descriptive Statistics	Value
Mean	183603.36
Median	182246.96
Minimum	78724.00
Maximum	255612.00
Range	176888.00

Standard Deviation	60102.51
Coefficient of Variation (%)	32.73

Table 1: Descriptive Statistics of Monthly Procurement Expenditure

2.3.3 Inventory Value and Concentration Analysis

Business problem: A substantial amount of working capital may be concentrated in a limited number of inventory categories. Identifying these categories is important for prioritising inventory monitoring and control.

Data used: Godown Summary containing inventory categories, quantities, and inventory values.

Method: Inventory categories were ranked in descending order of inventory value. The contribution of each category to total inventory value was calculated and used to identify the highest-value inventory categories. A ranking-based chart was prepared to visualise the concentration of inventory value among the major categories.

Formulae Used: Value Contribution (%) = (Inventory Value of Category / Total Inventory Value) × 100

Cumulative Contribution (%) = Sum of Value Contributions up to the respective category

Tool: Google Sheets.

Purpose: To identify high-value inventory categories and assess the extent to which inventory investment is concentrated among a limited number of items.

Measure	Result
Total Classified Inventory Categories	126.00
Total Classified Inventory Value	2669091.36
Number of top-value categories considered	5.00
Top 5 inventory value	1528793.05
Top 5 contribution in the total inventory value	57.28
Top 5 categories as % of total categories	3.97%

Table 2: Top Inventory Categories by Inventory Value

2.3.4 ABC Inventory Classification

Business problem: Inventory categories differ considerably in their contribution to total inventory value. Treating all categories with the same level of control may therefore lead to inefficient allocation of inventory-management effort.

Data used: Godown Summary.

Method: Inventory categories were sorted in descending order of inventory value.

Each category's percentage contribution to total inventory value was calculated, followed by the cumulative percentage contribution. Based on the cumulative inventory-value contribution, the categories were classified into A, B, and C groups according to the classification thresholds adopted for the study. The resulting classification was represented using an ABC/Pareto chart.

Categories were classified using cumulative inventory-value thresholds of approximately 70% for Class A, 70–90% for Class B, and above 90% for Class C

Tool: Google Sheets.

Purpose: To identify inventory categories requiring different levels of monitoring and control and to provide a quantitative basis for inventory-management recommendations.

ABC Class	No. of Items	Inventory Value	% of Total Inventory Value
A	8	1856441.35	69.55%
B	13	544976.18	20.42%
C	105	267673.83	10.03%
Total	126	2669091.36	100%

Table 3: Summary of ABC Inventory Classification

2.3.5 Workforce Attendance Analysis

Business problem: Workforce availability and attendance disruptions may affect the organisation's ability to maintain service operations efficiently.

Data used: Daily Attendance dataset covering April–June 2026, containing 936 employee-day attendance records for 12 employees.

Method: Attendance records were analysed to calculate relevant attendance and workforce-disruption indicators. The analysis examines attendance status, unplanned absence, late attendance, and other relevant disruption indicators available in the dataset. Summary tables and visualisations were used to identify patterns across the available workforce records.

Tool: Google Sheets.

Purpose: To identify patterns in workforce availability and attendance disruption that may be relevant to workforce planning and service-resource allocation.

2.3.6 Service Operations Analysis

Business problem: Variations in service workload and operational performance may affect service reliability, customer experience, and resource utilisation.

Data used: Service Operations dataset covering **April–June 2026**, containing **360 service-visit records**.

Method: Service records were analysed using relevant operational indicators available in the dataset, including service volume, SLA performance, service delays, first-time fix performance, customer complaints, backup deployment, and labour-cost measures. Summary statistics and visualisations were used to identify operational patterns and differences across the available service records.

Formulae Used:

First-Time Fix Rate (%) = (First-Time Fix Visits / Total Service Visits) × 100

Repeat Visit Rate (%) = (Repeat Visits / Total Service Visits) × 100

Complaint Rate (%) = (Complaint Visits / Total Service Visits) × 100

Complaint-Free Rate (%) = (Complaint-Free Visits / Total Service Visits) × 100

Tool: Google Sheets.

Purpose: To assess service-operation performance and identify areas requiring greater attention in service planning and resource utilisation.

Variable	Value
First-Time Fix = Y	321.00
First-Time Fix = N	39.00
Total	360.00
First-Time Fix Rate	89.17
Repeat Visit Rate	10.83
Complaint = Y	29.00
Complaint = N	331.00
Complaint Rate	8.06
Complaint-Free Rate	91.94
Average Rating	4.42
Median Rating	4.50

Minimum Rating	1.50
Maximum Rating	5.00
SD of Ratings	0.67

Table 4: Service Quality Performance Indicators

2.3.7 Workforce and Service-Operations Relationship Analysis

Business problem: Workforce disruptions may be associated with changes in service performance and resource requirements.

Data used: Daily Attendance and Service Operations datasets covering April–June 2026, containing 936 employee-day attendance records for 12 employees and 360 service-visit records.

Method: Service outcomes were compared across different workforce attendance-status categories. Indicators such as SLA compliance, service delays, backup deployment, and labour cost were examined to identify differences in operational outcomes across attendance conditions. The analysis is interpreted as a descriptive comparison and does not establish a causal relationship.

Formulae Used: $\text{SLA Compliance (\%)} = (\text{SLA Met Visits} / \text{Total Service Visits}) \times 100$

Tool: Google Sheets.

Purpose: To examine whether workforce availability patterns are associated with differences in service-operation outcomes and to identify areas for potential operational improvement.

Attendance Status	SLA Met	SLA Breached	Total Service Visits	SLA Compliance %
Present	263	18	281	93.59%
Late	23	11	34	67.65%
Half-day	13	0	13	100.00%
Absent - Unplanned	11	13	24	45.83%
Absent - Approved	5	3	8	62.50%
Total	315	45	360	87.50%

Table 5: SLA Compliance by Workforce Attendance Status

2.3.8 Credit Transaction Analysis

Business problem: Changes in business transaction activity provide useful context for understanding procurement and inventory-related operational patterns.

Data used: Credit Transaction Register containing monthly credit transaction values for January–June 2026.

Method: Monthly credit transaction values were summarised and visualised chronologically to identify changes in business activity during the study period. Summary measures were used to assess the magnitude and variability of monthly transaction activity.

Tool: Google Sheets.

Purpose: To understand variations in business activity and provide supporting context for interpreting procurement and inventory patterns.

Measure	Value
Number of Months	6.00
Mean	592032.27
Median	552415.75
Maximum	747928.82
Minimum	506224.30
Range	241704.52
Standard Deviation	94943.24
Coefficient of Variation	16.04

Table 6: Descriptive Statistics of Monthly Credit Transactions

2.3.9 Procurement–Credit Transaction Association

Business problem: Procurement requirements may vary with changes in business transaction activity. Examining the association between the two variables can provide preliminary insight into whether higher transaction activity coincides with higher procurement expenditure.

Data used: Purchase Register and Credit Transaction Register.

Method: Monthly procurement expenditure and credit transaction values were compared using a scatter plot and Pearson correlation coefficient. Because the current analysis contains only six monthly observations, the resulting relationship is interpreted as descriptive evidence of association rather than as a statistically conclusive or causal relationship.

Pearson Correlation Coefficient:

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{[\sum(x_i - \bar{x})^2 \times \sum(y_i - \bar{y})^2]}}$$

Coefficient of Determination: $R^2 = r^2$

Tool: Google Sheets.

Purpose: To examine the observed relationship between business transaction activity and procurement expenditure and provide additional context for procurement planning.

Month	Monthly Procurement	Credit Transaction Amount (₹)
January	178890.42	557,477.00
February	185603.50	525,287.00
March	226066.26	547,354.50
April	78724.00	667,922.00
May	255612.00	747,928.82
June	176724.00	506,224.30

Table 7: Monthly Procurement and Credit Transaction Values

Pearson Correlation (r): 0.07

Coefficient of Determination (R^2): 0.49%

3. Results and Findings

3.1 Procurement Trend and Variability Analysis

Procurement expenditure during January–June 2026 exhibited noticeable variability relative to its average monthly level. The standard deviation was ₹60,102.51, while the coefficient of variation was 32.73%. This indicates that monthly procurement expenditure varied substantially around the average during the study period. The result suggests that procurement requirements were not evenly distributed across the six-month period and provides a basis for examining procurement planning and inventory replenishment practices.

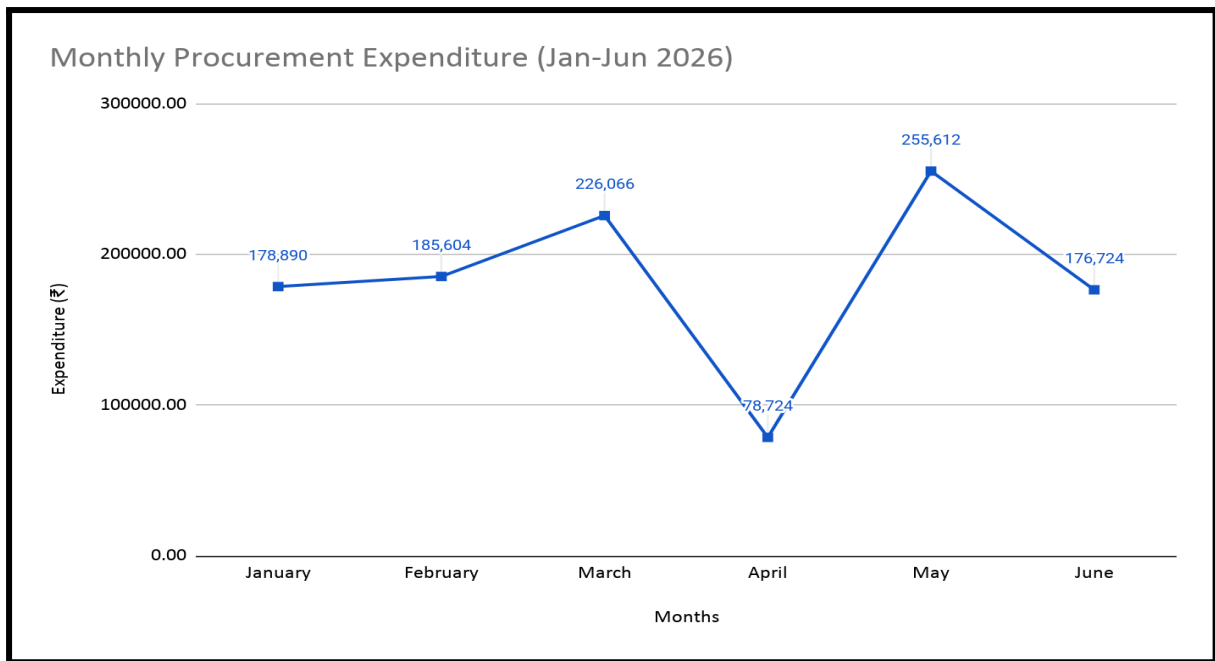


Figure 1: Monthly Procurement Expenditure Trend

3.2 Inventory Value and Concentration Analysis

The inventory value concentration analysis shows that a small number of categories account for a substantial proportion of the classified inventory value. The five highest-value categories, PRINTER, Ink, PRINTER REPAIRING, BROTHER DRUM 021, and TVSE RIBBON SPEED 40 PLUS, collectively account for ₹15.29 lakh, representing 57.28% of the total classified inventory value. These categories constitute only 3.97% of the 126 positive inventory records. This indicates a substantial concentration of inventory investment among a small group of high-value categories, highlighting the importance of prioritising these categories in inventory monitoring and control.

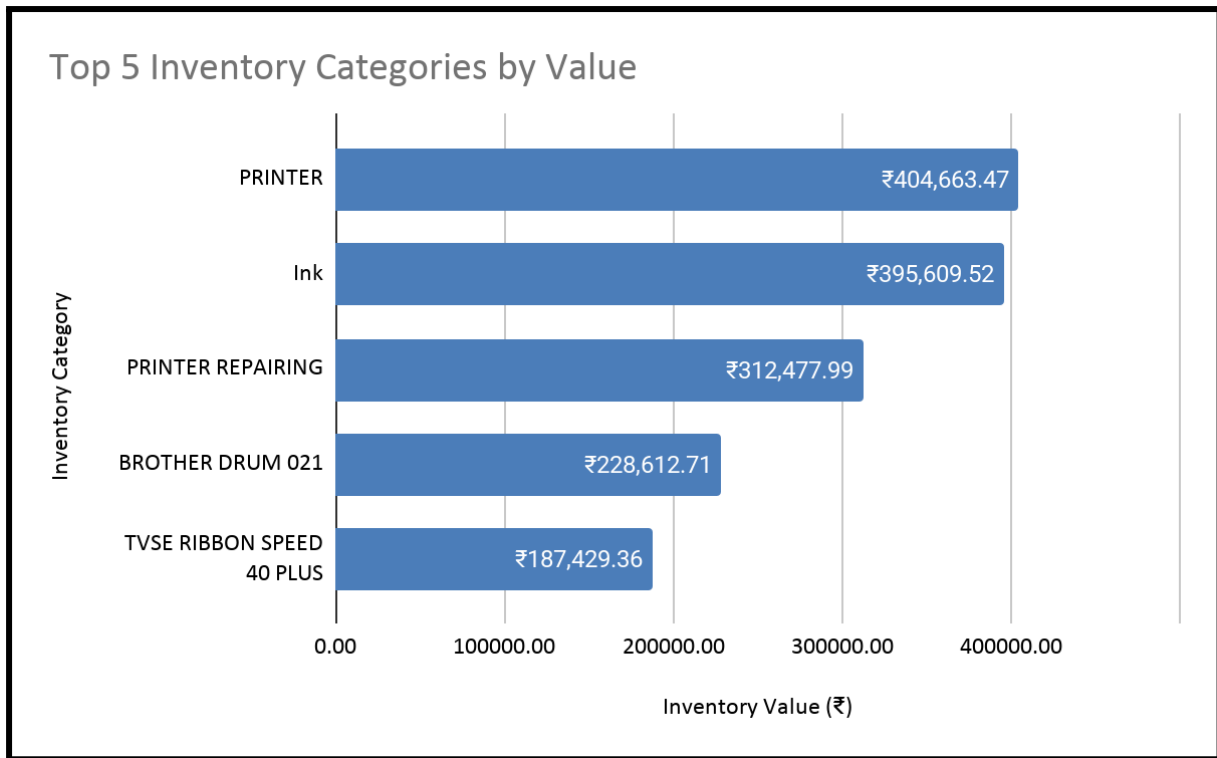


Figure 2: Top 5 Inventory Categories by Value

3.3 ABC Inventory Classification

ABC analysis was performed on the 126 inventory records representing positive inventory holdings in the Godown Summary. The inventory categories were ranked according to inventory value, followed by calculation of their individual and cumulative contribution to total inventory value. Based on the cumulative value thresholds, 8 categories were classified as A, 13 as B, and 105 as C.

The results indicate a high concentration of inventory value among a small number of categories. The 8 A-class categories represent only 6.35% of the classified inventory categories but account for 69.55% of total inventory value. The A-class inventory is primarily composed of printers, ink, printer-repairing items, drums, ribbons, paper, and cartridge-refilling categories. The first six A-class categories alone contribute 63.58% of total inventory value.

This concentration indicates that a relatively small group of inventory categories accounts for a substantial proportion of the capital invested in inventory. The ABC classification therefore provides a basis for prioritising these high-value categories in subsequent inventory-control and optimisation analysis.

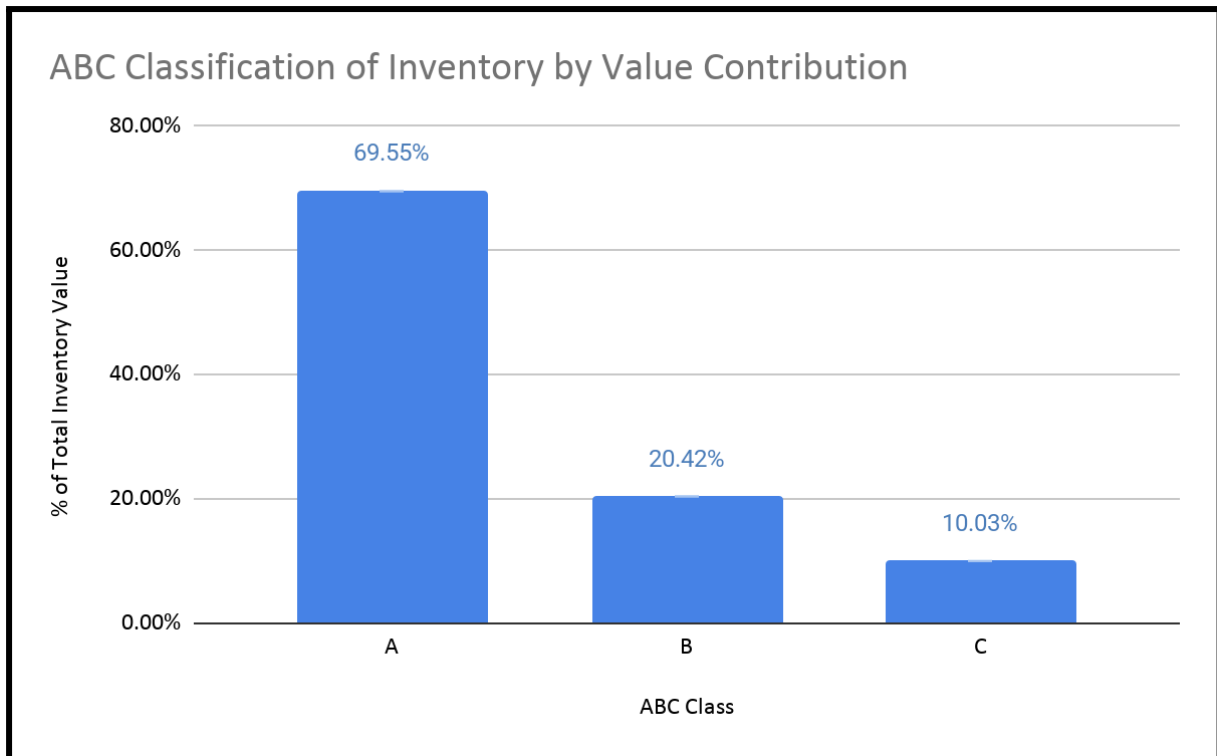


Figure 3: Inventory Value Distribution Across ABC Classes

3.4 Workforce Attendance vs Service Performance

SLA compliance varied across employee attendance categories. Service visits associated with **Present** employees achieved a compliance rate of **93.59%**, while **Late** attendance was associated with a lower compliance rate of **67.65%**. The lowest compliance was observed for **Absent - Unplanned** employees at **45.83%**, followed by **Absent - Approved** employees at **62.50%**. Although the **Half-day** category recorded **100.00%** compliance, it represented only **13 service visits**.

The observed pattern suggests that employee availability and attendance status may be associated with differences in service-level performance, particularly where **unplanned absence** occurs. However, the analysis is descriptive and does not establish a causal relationship between attendance status and SLA performance.

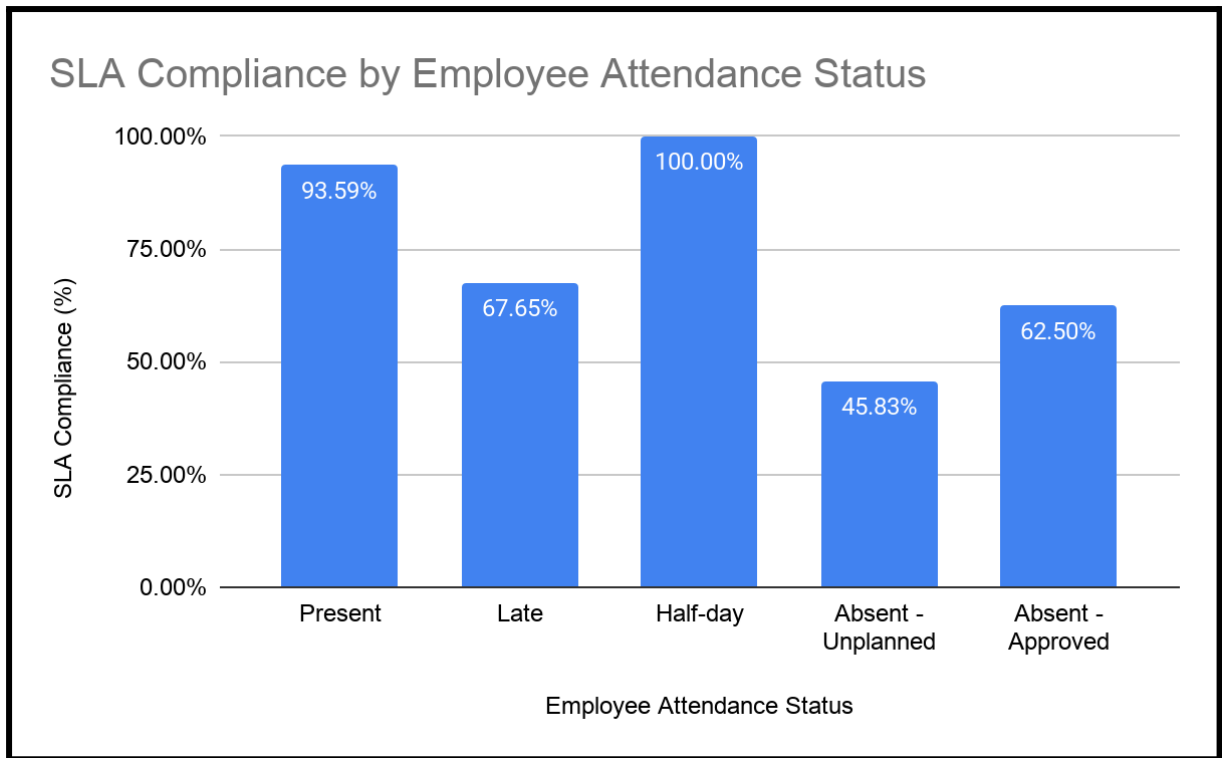


Figure 4: SLA Compliance Across Workforce Attendance Categories

3.5 Service Operations Analysis

The service-operations analysis indicates generally positive service performance during the study period. The First-Time Fix Rate was 89.17%, meaning that 10.83% of service visits required a repeat visit. The recorded customer complaint rate was 8.06%, while 91.94% of visits were complaint-free. Customer ratings averaged 4.42 out of 5, with a median of 4.50. Together, these indicators suggest generally satisfactory service delivery, while the proportion of repeat visits and recorded complaints identifies areas where service efficiency and quality can potentially be improved.

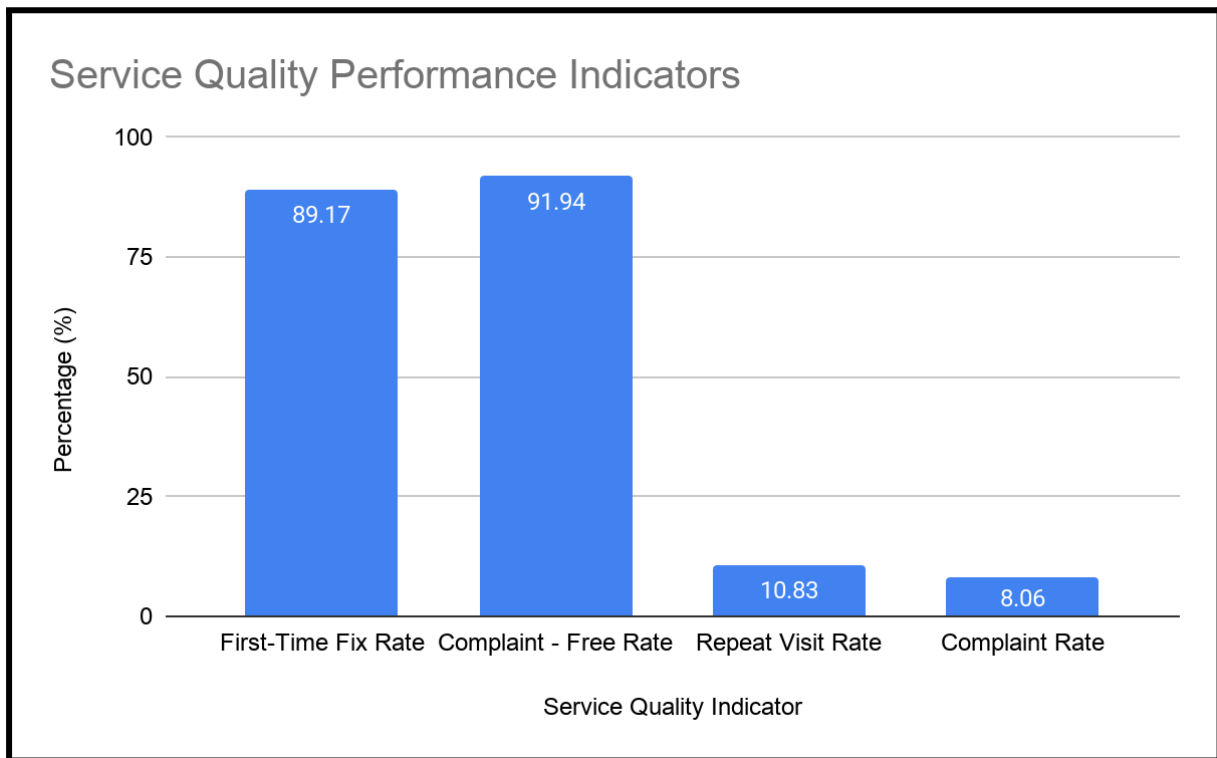


Figure 5: Service Quality Performance Indicators

3.6 Credit Transaction Analysis

The analysis of monthly credit transactions shows an average monthly credit transaction value of approximately ₹5.92 lakh, with a standard deviation of ₹0.95 lakh. The coefficient of variation was 16.04%, indicating moderate variation in monthly credit transaction activity during the six-month period. Credit transactions reached their highest level in May at approximately ₹7.48 lakh and their lowest level in June at approximately ₹5.06 lakh, resulting in a range of approximately ₹2.42 lakh. The observed variation indicates that credit-based business activity changes across months and should therefore be considered while interpreting transaction and working-capital patterns.

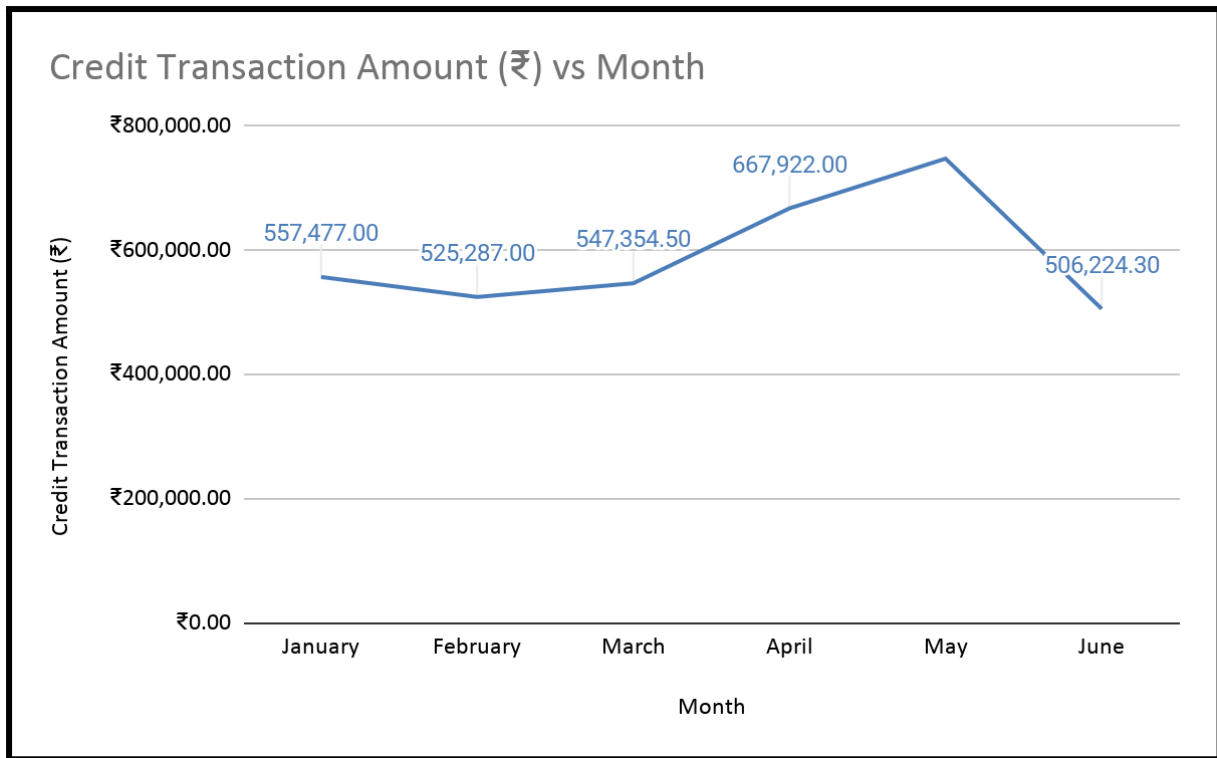


Figure 6: Monthly Credit Transaction Trend

3.7 Procurement–Credit Transaction Association

The Pearson correlation between monthly credit transaction value and procurement expenditure was approximately 0.07, indicating a very weak positive linear association during the January–June 2026 study period. The corresponding R^2 of approximately 0.49% indicates that very little of the observed variation in monthly procurement expenditure is explained by its linear association with credit transaction value. This suggests that procurement expenditure during the study period may have been influenced by factors other than the level of recorded credit transactions.

However, this analysis is based on only six monthly observations and should therefore be interpreted as preliminary descriptive evidence rather than a statistically conclusive relationship.

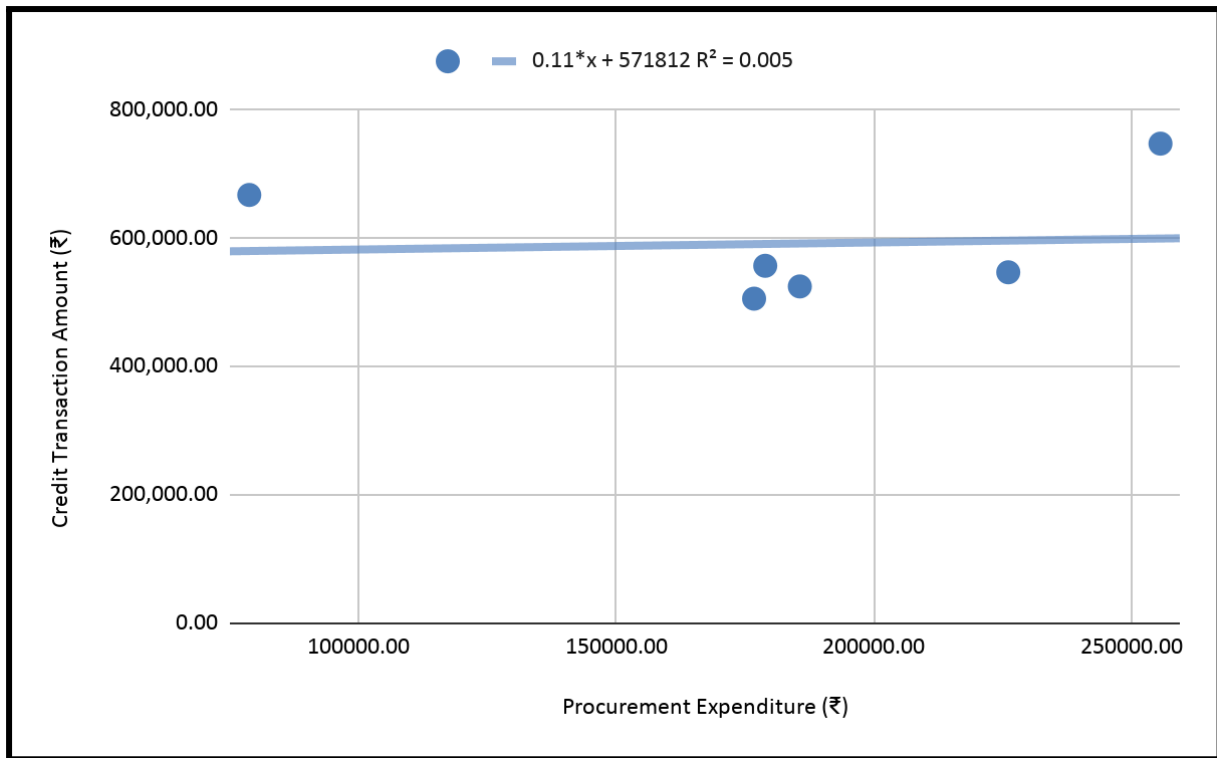


Figure 7: Relationship Between Procurement Expenditure and Credit Transactions

4. Interpretation of Results

4.1 Inventory Concentration and Control

The inventory analysis indicates that inventory investment is highly concentrated among a relatively small number of categories. The five highest-value inventory categories account for approximately ₹15.29 lakh, or 57.28% of the total classified inventory value, while representing only 3.97% of the 126 positive inventory records. The ABC analysis provides further evidence of this concentration, with only eight A-class categories accounting for 69.55% of total inventory value.

This indicates that inventory-management effort does not need to be distributed equally across all categories. A relatively small number of high-value items represents a substantial share of the capital invested in inventory. Ineffective control of these categories could therefore have a disproportionately large impact on working-capital utilisation and inventory efficiency.

The results suggest that A-class and other high-value inventory categories should receive greater managerial attention. Regular review of stock levels, procurement requirements, and inventory movement for these categories can help reduce unnecessary capital being held in

stock. Lower-value C-class categories can continue to be monitored using comparatively simpler control procedures.

4.2 Procurement Planning and Variability

Monthly procurement expenditure during January–June 2026 showed considerable variation. The standard deviation was **₹60,102.51**, with a coefficient of variation of approximately **32.73%**. Procurement expenditure ranged from **₹78,724 in April to ₹2,55,612 in May**, indicating that purchasing requirements were not evenly distributed across the study period. The variation suggests that procurement activity may be influenced by changing inventory requirements and operational needs across months. The large difference between the April and May expenditure levels particularly indicates that procurement decisions can vary substantially over a short period.

However, the available six-month dataset does not provide sufficient evidence to determine the exact causes of these fluctuations. Therefore, the result should primarily be used as an indicator that procurement planning requires closer monitoring rather than as evidence of a specific underlying cause.

4.3 Credit Transaction Activity and Business Implications

Credit transaction activity also varied during the study period. The average monthly credit transaction value was approximately ₹5.92 lakh, with a standard deviation of approximately ₹0.95 lakh and a coefficient of variation of 16.04%. Credit transactions reached their highest level in May at approximately ₹7.48 lakh and their lowest level in June at approximately ₹5.06 lakh.

The relatively lower coefficient of variation compared with procurement expenditure indicates that credit transaction activity was comparatively more stable over the six-month period. Nevertheless, the difference between the highest and lowest monthly values indicates that business activity was not completely uniform.

These fluctuations provide useful context when interpreting procurement and inventory patterns. However, credit transaction activity should not automatically be treated as a direct indicator of procurement requirements, particularly because the procurement–credit analysis found only a very weak linear association.

4.4 Procurement–Credit Transaction Relationship

The relationship analysis produced a Pearson correlation coefficient of approximately **0.07**, with an R² of approximately **0.49%**. This indicates a very weak positive linear association between monthly credit transaction values and procurement expenditure during the study

period. In practical terms, the observed variation in procurement expenditure was largely not explained by the linear relationship with recorded credit transaction values.

This finding suggests that procurement decisions during the study period were likely influenced by factors beyond the recorded credit transaction amount. These may include inventory requirements, specific customer orders, supplier-related factors, or operational requirements; however, the available data does not allow these factors to be established directly.

Since the analysis is based on only **six monthly observations**, the result should be treated as preliminary descriptive evidence rather than a statistically conclusive relationship.

4.5 Workforce Availability and Service Performance

The workforce–service analysis shows noticeable differences in SLA compliance across employee attendance categories. Service visits associated with Present employees recorded 93.59% SLA compliance, compared with 67.65% for Late attendance. The lowest compliance was observed for Absent - Unplanned employees at 45.83%, while Absent - Approved employees recorded 62.50%. Although the Half-day category recorded 100.00% compliance, it represented only 13 service visits.

The pattern suggests that workforce availability and attendance status may be associated with service-level performance. In particular, the substantially lower SLA compliance associated with unplanned absence indicates that unexpected workforce shortages may create difficulties in maintaining service commitments. However, this result should be interpreted as descriptive evidence of an observed association rather than a causal relationship. The differences in SLA compliance may also reflect other operational factors not captured in the available data.

4.6 Service Operations and Quality

The service-operation analysis indicates generally satisfactory service performance during the study period. The **First-Time Fix Rate was 89.17%**, while the **complaint-free rate was 91.94%**. Customer ratings averaged **4.42 out of 5**, with a median rating of **4.50**.

These indicators suggest that the overall service operation is performing reasonably well. At the same time, the **10.83% repeat-visit rate** indicates that a proportion of service cases require additional visits. Similarly, the **8.06% complaint rate** identifies a measurable area where service quality could potentially be improved.

Rather than indicating a major service-quality problem, these results identify opportunities for **incremental improvement in service efficiency, first-time resolution, and complaint reduction**.

5. Recommendations and Limitations

5.1 Recommendations

Based on the analysis, the following recommendations are proposed for Ashmiti Enterprise:

5.1.1 Prioritised Inventory Control

The ABC analysis reveals that a small number of items drive most inventory value (e.g., 8 A-class categories account for 69.55%).

Recommendation: Apply stricter inventory monitoring and more frequent reviews to A-class items. Simpler control procedures suffice for B- and C-class categories.

5.1.2 Improve Procurement Planning

Procurement expenditure showed considerable variation (coefficient of variation: **32.73%**) during the study.

Recommendation: Implement regular monthly procurement reviews that account for inventory requirements, past spending, and ABC classification to avoid unnecessary capital tie-up.

5.1.3 Strengthen Workforce Contingency Planning

Unplanned absence was associated with substantially lower SLA compliance (45.83%) compared with present employees (93.59%).

Recommendation: Develop a contingency mechanism for workforce shortages, such as maintaining backup personnel and prioritising critical service visits.

5.1.4 Improve First-Time Service Resolution

The repeat-visit rate was **10.83%**, indicating that a proportion of service cases required additional visits.

Recommendation: Systematically record reasons for repeat visits to identify recurring issues. Use this data to improve pre-visit diagnosis, spare-parts preparation, and technician allocation.

5.1.5 Monitor Customer Complaints

8.06% of service visits resulted in recorded customer complaints.

Recommendation: Categorise and periodically review customer complaints to identify root causes and focus corrective actions on high-impact service issues.

5.1.6 Use Credit Transactions as a Supporting Indicator

Procurement expenditure shows a very weak linear relationship with credit transaction activity (correlation: ~ 0.07).

Recommendation: Use credit transaction activity only as a supporting indicator. Base procurement decisions on inventory levels, ABC classification, and operational requirements instead.

5.2 Limitations

The findings and recommendations of the project should be interpreted in light of the following limitations.

5.2.1 Limited Study Period

The business transaction analysis covers January to June 2026, providing only six months of observations. This limits the ability to identify longer-term seasonal patterns or establish whether the observed procurement and credit trends persist over a longer period.

5.2.2 Workforce and Service-Operations Data

The workforce and service-operation analysis covers a three-month period from April to June 2026 and was conducted to examine workforce availability, service performance, and operational outcomes. Therefore, the findings should be interpreted as indicative operational patterns within the study period and should be considered alongside other operational factors when evaluating workforce and service performance.

5.2.3 Limited Observations for Correlation Analysis

The procurement–credit relationship was analysed using six monthly observations. Although the analysis provides an indication of the relationship within the available data, the small number of observations limits the strength of conclusions that can be drawn from the correlation and R^2 values.

5.2.4 Availability of Business Records

The analysis was restricted to the business records that were made available for the project. Consequently, certain operational factors that may influence procurement, inventory, credit transactions, or service performance could not be incorporated into the analysis.

5.2.5 Descriptive Nature of the Analysis

The analyses primarily identify patterns, variation, concentration, and associations within the available datasets. They should therefore be interpreted as decision-support findings rather than causal evidence or definitive forecasts.

6. Additional

Supporting Documentation and Links:  **BDM Capstone Project**